L26

L26_Stochastic Processes_...
PDF

Generate a lecture scribe intended strictly for exam preparation for the course
CSE 400: Fundamentals of Probability in Computing (School of engineering and applied sciences, Ahmedabad University).
Strictly use only the provided context, which consists of:
1. Basics of Stochastic Processes [Revision]
– Concept of Waveform Collections and Realizations
– Process Characterization, First, Second and Nth Order Density Functions
2. Statistical Moments:
– Mean Vector and Autocorrelation Function
– Autocorrelation Properties, Positive Definite Matrices and Proofs
3. Stationary Stochastic Process:
– Strict Sense Stationary (SSS) Process, Wide Sense Stationary (WSS) Process
– HW Example: White Noise Process
4. Ergodic Process:
– Time Average of a Random Process
Do not introduce any concepts, results, derivations, explanations, interpretations, or examples that do not appear in the provided context.
While writing the lecture scribe, reason step by step and include the reasoning explicitly in the scribe for all of the topics in order of their appearance in PPT slides only when they appear in the context:
Each derivation, computation, or example must be written step by step, without skipping intermediate steps, exactly as reflected in the context.
Maintain the authenticity of mathematical equations.
Organize the scribe using clear section headings and logical flow, so that a student can rely on this document alone for closed-notes exam revision.
Do not:
- Add intuition, motivation, or commentary not stated in the lecture

- Simplify or reinterpret steps
- Introduce alternative methods or explanations
- Add new examples, exercises, or conclusions

Important Note - Strictly follow the instructions as mentioned above.

Below is a **strict lecture scribe** constructed **only from the provided slides** .
All steps, equations, and reasoning are written **exactly in sequence and form as they appear**, without any additions.

CSE 400: Fundamentals of Probability in Computing

Lecture 26: Stochastic Processes (SSS, WSS, Ergodic)

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1. Basics of Stochastic Processes [Revision]

1.1 Collection of Waveforms

Stochastic Process Definition

A stochastic process $X(\zeta, t)$ is an entire collection of waveforms or functions over time dependent on the outcome of a random experiment.

- The random output heavily depends on the experiment.

Example

Tossing a coin gives heads or tails resulting in specific deterministic waveforms.

1.2 Realizations and Random Variables

Fixed in Time Representation

Step 1:
A specific measured waveform from the experiment is a **Realization** of the process.

Step 2:
If we freeze the process at a specific time $t = t_1$, the process output evaluates to a single Random Variable:

$$X(\zeta, t_1) = X(t_1)$$

Step 3:
By evaluating at multiple time instances (t_1, t_2), we acquire multiple dependent random variables:

$$X(t_1), X(t_2)$$

Step 4:
The dependency between these generated random variables dictates the fundamental nature of the stochastic process.

1.3 Characterization of Stochastic Process

Probability Density Functions (PDF)

Step 1: 1st Order Characterization

Describes the random variable at a single time t_1:

$$t_1 \to f_{X_1}(x, t_1) \to F_{X_1}(x, t_1)$$

Step 2: 2nd Order Characterization

Describes joint probability across t_1 and t_2:

$$(t_1, t_2) \to f_{X_1 X_2}(x_1, x_2, t_1, t_2)$$

Step 3: nth Order Characterization

Complete statistical description:

$$(t_1, \ldots, t_n) \to f_{X_{1,} \ldots, X_n}(x_1, \ldots, x_n, t_1, \ldots, t_n)$$

2. Statistical Moments

2.1 Mean Vector (First Order Moment)

Mean Definition

Step 1: Definition

$$\mu_X(t) = E[X(t)]$$

Step 2: Computation using PDF

$$\mu_X(t) = \int_{-\infty}^{\infty} x \cdot f_X(x, t)\, dx$$

Step 3: Property

$$\mu_X(t) \text{ changes as a function of time}$$

2.2 Autocorrelation Function

Definition

Step 1: Definition

$$R_{XX}(t_1, t_2) = E[X(t_1)X^*(t_2)]$$

Step 2: Using Joint Density (2nd Order Characterization)

$$R_{XX}(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 \cdot f_{X_1 X_2}(x_1, x_2, t_1, t_2)\, dx_1 dx_2$$

2.3 Properties of Autocorrelation

Conjugate Symmetry

Step 1: Property

$$R_{XX}(t_1, t_2) = R_{XX}^*(t_2, t_1)$$

Step 2: Proof Sketch

$$R_{XX}^*(t_2, t_1) = (E[X(t_2)X^*(t_1)])^*$$

$$(E[X(t_2)X^*(t_1)])^* = E[X^*(t_2)X(t_1)]$$

$$E[X^*(t_2)X(t_1)] = E[X(t_1)X^*(t_2)] = R_{XX}(t_1, t_2)$$

Positive Definiteness

Autocorrelation function forms a positive definite matrix.

2.4 Covariance Function

Definition

$$C_{XX}(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X^*(t_2) - \mu_X^*(t_2))]$$

Normalized Form

$$\rho_{XX} = \frac{C_{XX}(t_1, t_2)}{\sigma_X \sigma_X}$$

2.5 Positive Definite Function

Quadratic Form Definition

For arbitrary times $t_1, \ldots, t_n$ and constants $a_1, \ldots, a_n$:

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j^* R_{XX}(t_i, t_j) \geq 0$$

Matrix Expansion

$$[a_1^* \; a_2^* \; \ldots \; a_n^*] \begin{bmatrix} R_{11} & R_{12} & \ldots \\ R_{21} & R_{22} & \ldots \\ \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \end{bmatrix} > 0$$

Equivalent Form

$$a^H Ra > 0$$

2.6 Positive Definite Matrices

Conditions

1. A symmetric matrix is positive definite if all eigenvalues are positive.
2. A matrix is positive definite if all leading principal minors are positive:

$$|A_1| > 0, \; |A_2| > 0, \; \ldots, \; |A_n| > 0$$

Example (Step-by-Step)

Matrix:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Step 1: First Minor

$$|A_1| = 2 > 0$$

Step 2: Second Minor

$$|A_2| = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = (2)(2) - (-1)(-1) = 4 - 1 = 3 > 0$$

Step 3: Third Minor

$$|A_3| = 2(4-1) + 1(-2-0)$$

$$= 2(3) - 2 = 6 - 2 = 4 > 0$$

Final Result:

Matrix is positive definite.

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3. Stationary Stochastic Process

3.1 Strict Sense Stationary (SSS)

Step 1: First Order

$$f_X(x,t) = f_X(x, t+\tau) = g(x)$$

$$\mu_X(t) = \int x g(x)\, dx = \mu_X = \text{constant}$$

Step 2: Second Order

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 + \tau, t_2 + \tau)$$

Choose:

$$\tau = -t_2$$

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 - t_2, 0)$$

Step 3: Autocorrelation Derivation

$$R_{XX}(t_1,t_2) = \iint x_1x_2f_X(x_1,x_2;t_1,t_2)\,dx_1dx_2$$

$$= \iint x_1x_2f_X(x_1,x_2;t_1-t_2)\,dx_1dx_2$$

$$= \phi(t_1-t_2)$$

Final Result

$$R_{XX}(t_1,t_2) = R_{XX}(t_1-t_2)$$

3.2 Wide Sense Stationary (WSS)

Definition

SSS requires all orders to be time-invariant.
WSS is a weaker condition.

Condition 1: Constant Mean

$$E[X(t)] = \mu_X = \text{constant}, \quad \forall t$$

Condition 2: Autocorrelation depends only on lag

$$E[X(t_1)X^*(t_2)] = R_{XX}(\tau), \quad \tau = t_1 - t_2$$

3.3 Relationship

$$\text{SSS} \Rightarrow \text{WSS}, \quad \text{but} \quad \text{WSS} \not\Rightarrow \text{SSS}$$

Gaussian Process Exception

If $X(t)$ is Gaussian and WSS:

$$\text{WSS} \Rightarrow \text{SSS}$$

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4. Ergodic Process

Time Average of a Random Process

(Only heading provided in context; no further derivation present in slides.)

END OF SCRIBE

If you want, I can convert this into **last-day revision notes (1–2 pages ultra-condensed)** or **expected exam questions from this exact lecture**.